

British bacteriologist Alexander Fleming's accidental discovery of penicillin—the first antibiotic—was one of the greatest breakthroughs in the fight against disease. But it took another decade before other scientists managed to transform his research into a miracle drug that would save millions of lives.

MILESTONES

CHANCE FINDING

Discovers *Penicillium notatum* by accident in 1928, in an abandoned Petri dish in his laboratory.

KEY PUBLICATION

Publishes his findings in the *British Journal of Experimental Pathology* in 1929.

FIRST TREATMENT

Penicillin is trialed for the first time in 1941, on Albert Alexander, but he dies due to insufficient supplies.

WAR WOUNDED

Between 1943 and 1945, hundreds of thousands of war casualties are treated with penicillin.

JOINT AWARD

Awarded Nobel Prize in Physiology or Medicine jointly with Florey and Chain in 1945.

Penicillin was available as a treatment to many wounded soldiers during World War II. In 1944, it was used on troops in the D-Day landings, and had a huge impact on reducing the death toll.

Alexander Fleming was born into a poor farming family in Ayrshire, Scotland. At 16, he went to work for a shipping company, but 4 years later, he inherited some money from an uncle and decided to use it to study medicine. In 1906, he qualified with distinction at St. Mary's Hospital Medical School, London.

When World War I broke out, Fleming joined the Royal Army Medical Corps. Based at a military hospital in France, seeing first-hand the widespread deaths of soldiers from infected wounds deeply affected him. He realized that the antiseptics used were actually causing harm when applied to deep wounds. When he returned to his research post at St. Mary's at the end of the war, he resolved to study "bacteria killers." In 1923, he identified the natural enzyme lysozyme, which has antibacterial properties, but the research was of limited success.

**"I did not invent penicillin.
Nature did that."**

Alexander Fleming

